Dividend Omission Effects in Black-Scholes-Merton Pricing of Short-Maturity U.S. Equity Call Options

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Abstract

The effect of dividends on Black–Scholes–Merton (BSM) call values is theoretically established. We instead quantify the distortion caused by setting a positive dividend yield to zero and ask whether correcting that omission improves agreement with observed market prices. The sample contains short-maturity calls on nine U.S. equities observed on 13 trading days from August 29 through September 18, 2025, with a common September 19, 2025 expiration. To avoid circularity from option-implied volatility, the primary specification uses a fixed annualized volatility estimated independently from 20 pre-sample daily log returns ending August 28, 2025. Market premiums are valid bid–ask midpoints, and a uniform activity/quote screen retains 5,115 of 9,959

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option-day observations. The first-order dividend-sensitivity approximation to the exact BSM omission effect is extremely accurate (correlation 0.999996; mean absolute approximation error \$0.000086). Empirical pricing effects are heterogeneous: the dividend adjustment lowers MAE for UNH (4.58%) and BAC (3.89%), is economically negligible for AAPL and NVDA, and raises MAE for several other equities, including JPM, MRK, MSFT, and WFC. A Yahoo-implied-volatility diagnostic again produces near-perfect market–model correlations, illustrating why such correlations cannot be interpreted as independent model-validation evidence. The results separate the predictable theoretical effect of dividend omission from the empirical question of whether the dividend correction moves an otherwise misspecified benchmark closer to observed American-style option prices.

1 Introduction

The Black–Scholes framework remains a central benchmark for option valuation because of its tractability and transparent dependence on model inputs [1]. Merton's extension incorporates a continuous dividend yield, q, and establishes that expected dividend payments reduce the value of a European call relative to an otherwise identical non-dividend-paying underlying [2]. The sign of the dividend adjustment is therefore not an empirical discovery. The relevant empirical questions are how large the price distortion is when a positive dividend yield is set to zero and whether correcting that assumption materially improves agreement with observed option prices.

That distinction matters because market–model discrepancies are not generated by the dividend assumption alone. U.S. single-stock options are American-style, whereas the closed-form BSM formula is European; volatility is not constant in market data; dividends are discrete rather than continuous; and option quotes can be stale or wide. In addition, an implied volatility obtained from an option premium is mechanically linked to that premium. Reusing such an implied volatility in the same or a closely related pricing equation and interpreting the resulting price correlation as independent evidence of model accuracy creates a circularity problem.

This study therefore focuses on dividend omission within a controlled BSM comparison. We derive an analytical first-order benchmark for the effect of replacing $q > 0$ with $q = 0$, use volatility estimated independently from pre-sample stock returns for the primary empirical exercise, define the observed option premium explicitly from bid–ask quotes, apply uniform activity and no-arbitrage screens, and evaluate performance using both dollar and normalized error measures with paired daily tests. Yahoo Finance implied volatility is retained only as a diagnostic specification rather than as primary validation evidence.

2 Related Work

Black and Scholes [1] derived the closed-form European option-pricing relation under constant volatility, continuous trading, and no dividends, while Merton [2] incorporated a continuous proportional dividend yield. Research on dividend-paying American calls, including Roll [3], Geske [4], and Whaley [5], showed why dividends can create an early-exercise incentive. The binomial framework of Cox, Ross, and Rubinstein [6] provides a convenient numerical benchmark for this American feature.

Empirical studies have long documented systematic departures from constant-volatility BSM pricing. MacBeth and Merville [7] reported moneyness-related pricing errors, Rubinstein [8] documented cross-strike variation in implied volatility, and Dumas, Fleming, and Whaley [9] showed that richer volatility specifications can improve in-sample fit without guaranteeing superior out-of-sample performance. Stochastic-volatility models such as Hull and White [10] and Heston [11], as well as the empirical comparisons of Bakshi, Cao, and Chen [12], further demonstrate that volatility misspecification can be quantitatively important.

Short-dated options are useful for studying local pricing effects but are also sensitive to market microstructure and quote quality [13, 14]. Because spreads and transaction timing can materially affect empirical measurements, market-price construction must be documented explicitly; Muravyev and Pearson [15] emphasize the importance of measuring option trading costs carefully. Our contribution is intentionally narrow: we quantify the BSM distortion caused by imposing $q = 0$, compare the exact distortion with the option's dividend sensitivity, and then test whether the correction improves market-price accuracy once volatility and data-quality choices are separated from the observed option premium.

3 Methodology

3.1 Pricing equations and analytical omission effect

For a European call on a non-dividend-paying stock,

$$C_0 = SN(d_1) - Ke^{-rT}N(d_2), \tag{1}$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}. \tag{2}$$

With a continuous dividend yield q,

$$C_q = Se^{-qT}N(d_{1,q}) - Ke^{-rT}N(d_{2,q}), \tag{3}$$

where

$$d_{1,q} = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_{2,q} = d_{1,q} - \sigma\sqrt{T}. \tag{4}$$

The call's dividend-yield sensitivity is

$$\frac{\partial C_q}{\partial q} = -TSe^{-qT}N(d_{1,q}) < 0. \tag{5}$$

Therefore, a first-order approximation to the distortion generated by replacing $q > 0$ with zero is

$$C_0 - C_q \approx qTSe^{-qT}N(d_{1,q}). \tag{6}$$

This relation supplies an analytical benchmark; it does not by itself establish that C_q must be closer to the observed market premium than C_0.

3.2 Data, market premium, timing, and sample construction

The option-chain snapshots were collected from Yahoo Finance for nine U.S. equities: AAPL, BAC, JPM, MRK, MSFT, NVDA, PFE, UNH, and WFC. The 13 observation dates are August 29 and September 2–4, 8–12, and 15–18, 2025. All contracts in the analysis expire September 19, 2025. The raw files contain 9,959 option-day observations.

The primary observed premium is the bid–ask midpoint, $C^{mkt} = (Bid + Ask)/2$, rather than the last transaction price. We require strictly positive two-sided quotes, $Ask \geq Bid$, positive stock and strike prices, and the call upper bound $C^{mkt} \leq S$. Because the downloaded files do not contain a separate timestamp for the displayed bid and ask, exact intraday synchronization cannot be verified. We therefore use `lastTradeDate` as an activity screen, not as a quote timestamp: the primary sample requires a transaction on the snapshot date or no more than one preceding U.S. trading session. The same rules are applied to every ticker. This leaves 5,115 observations. Robustness checks vary the activity threshold and cap relative bid–ask spreads.

The raw audit confirms that freshness screening is material. In particular, NVDA contains 1,599 observations with strikes above 200, of which 1,092 have last-trade dates before June 10, 2024. Those legacy records contribute to the extreme strike range and implied-volatility values visible in the unfiltered data. No ticker-specific deletion is used; the common quote, no-arbitrage, and activity screens determine the retained sample.

Time to maturity is recomputed using an ACT/ACT convention. Because all observations and the expiration occur in non-leap year 2025, this is actual calendar days to expiration divided by 365. The risk-free rate is held fixed at $r = 0.0441$ (4.41%) across the short sample, matching the value used in the original computational workflow. Dividend yields are the ticker-level Yahoo Finance yields stored with the collected data.

3.3 Independent volatility input

The primary volatility parameter is estimated independently of option premiums. For each equity, we use 21 daily closing prices from July 31 through August 28, 2025, which produce 20 pre-sample log returns,

$$r_t = \ln(S_t/S_{t-1}), \qquad \widehat{\sigma}_{20} = \sqrt{252}\, sd(r_1, \ldots, r_{20}). \tag{7}$$

The resulting volatility estimate is fixed across the 13-day option sample. This preserves independence from option prices, avoids look-ahead within the sample, and is consistent with the constant-volatility benchmark being evaluated. Ten- and fifteen-return windows are reported as robustness checks.

3.4 Error measures and inference

For each retained contract,

$$AE_{0,i} = |C_{0,i} - C_i^{mkt}|, \qquad AE_{q,i} = |C_{q,i} - C_i^{mkt}|. \tag{8}$$

The contract-level improvement is $AE_{0,i} - AE_{q,i}$, so positive values indicate lower error after incorporating dividends. Ticker-level mean absolute error (MAE) remains useful in dollars, but cross-stock comparisons use normalized MAE,

$$NMAE_j = \frac{\sum_{i \in j} |C_i^{model} - C_i^{mkt}|}{\sum_{i \in j} C_i^{mkt}}. \tag{9}$$

To avoid treating multiple strikes on the same day as independent replicates, absolute errors are first averaged within ticker and date, yielding 13 paired daily MAEs per ticker. We report paired t-tests and two-sided Wilcoxon signed-rank tests.

3.5 American-style diagnostic and implied-volatility diagnostic

The observed U.S. single-stock calls are American-style, whereas Equations (1)–(4) value European calls. We therefore interpret $C_0 - C_q$ strictly as the dividend-omission effect within a common European framework. The residual $C^{mkt} - C_q$ may additionally contain an early-exercise premium, discrete-dividend effects, volatility misspecification, and microstructure noise. To quantify the exercise-style component, we price each retained contract with a 100-step Cox–Ross–Rubinstein tree using the same q and independent volatility and compare its American and European tree values [6].

As a separate diagnostic, we repeat the two BSM calculations using Yahoo Finance implied volatility while retaining the revised sample and market-price rules. Because implied volatility contains option-price information, this specification is not treated as independent model-validation evidence.

4 Empirical Results

4.1 Cleaned sample

Table 1 summarizes the final primary sample. The filters are intentionally ticker-neutral. The remaining mean last-trade age is below one calendar day for every ticker, while the mean relative spread varies substantially across underlyings.

Table 1: Sample construction and primary inputs by ticker

Ticker	Raw N	Primary N	Mean age	Mean rel. spread	Avg. midpoint	HV20	Div. yield
AAPL	1061	654	0.15	7.1	55.89	29.3	0.45%
BAC	708	388	0.30	8.2	6.15	23.9	2.21%
JPM	839	465	0.25	9.4	36.94	17.2	1.86%
MRK	620	270	0.23	22.9	5.33	20.9	3.85%
MSFT	1304	786	0.21	14.3	51.40	16.4	0.66%
NVDA	3037	1336	0.19	8.0	51.05	23.7	0.02%
PFE	477	189	0.33	20.1	2.13	28.4	6.95%
UNH	1276	700	0.21	24.6	38.81	52.0	2.85%
WFC	637	327	0.24	10.4	6.44	25.1	2.19%

Notes: Mean age is the calendar-day difference between the snapshot date and `lastTradeDate` within the activity-screened sample. Mean relative spread is $100(Ask - Bid)/Midpoint$. HV20 and dividend yield are annual percentages.

Figure 1 replaces date-specific raw-strike plots by pooling all 13 dates and normalizing strike and option value by the contemporaneous stock price on common panel scales.

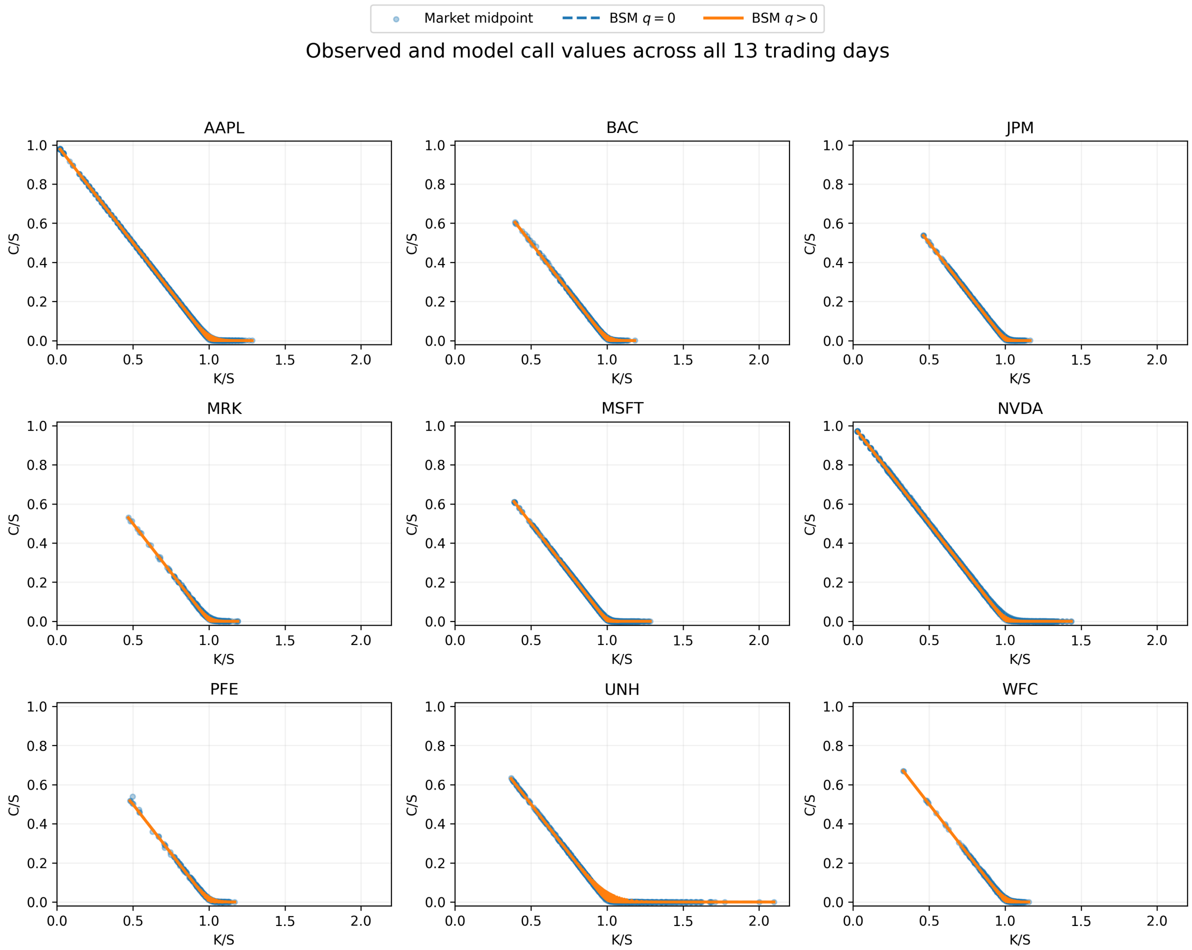

Figure 1: Observed bid–ask midpoints and model values across all 13 trading days. Each panel is one underlying; strike and call value are normalized by the contemporaneous stock price. All panels use a common K/S range of 0–2.2 and a common C/S range.

4.2 Mechanical dividend-omission effect

Across the 5,115 retained observations, Equation (6) reproduces the exact model difference $C_0 - C_q$ with correlation 0.999996. The mean exact omission effect is \$0.0422 per option, the mean absolute approximation error is \$0.000086, and the maximum absolute approximation error is \$0.0018. Thus, at these short maturities, the local dividend sensitivity is an accurate approximation to the mechanical effect of setting q to zero.

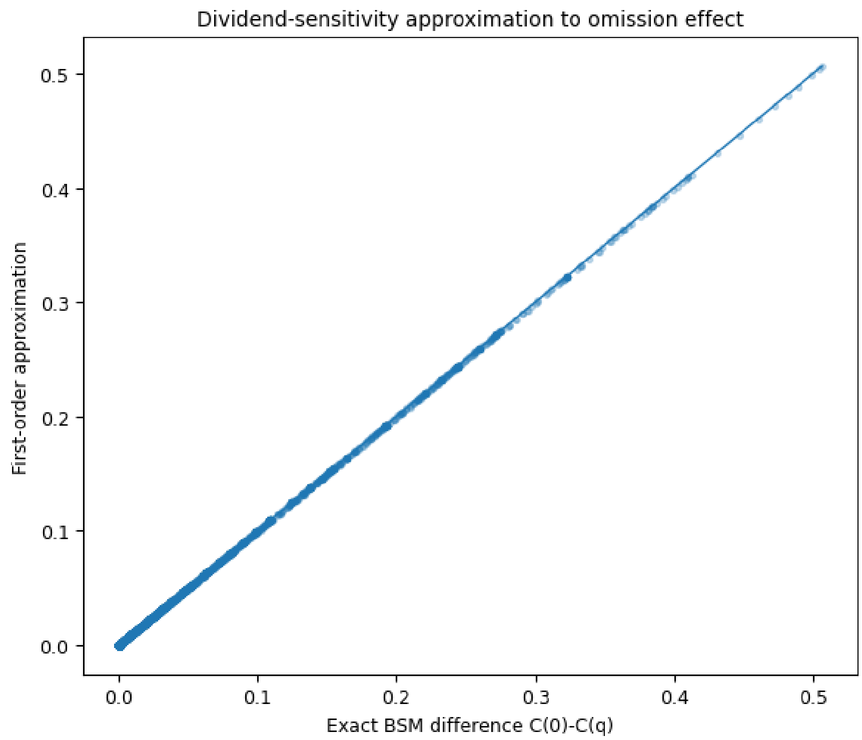

Figure 2: Exact dividend-omission effect $C_0 - C_q$ versus the first-order approximation $qTSe^{-qT}N(d_{1,q})$.

4.3 Market-price accuracy under independent volatility

Table 2 shows that the theoretical direction of the dividend correction does not imply a uniform reduction in market-pricing error. UNH exhibits a pooled MAE improvement of 4.58%, and BAC improves by 3.89%. AAPL and NVDA are essentially unchanged in economic terms. In contrast, the dividend-adjusted European BSM value has larger MAE for JPM, MRK, MSFT, PFE, and WFC. The direction can be understood from signed errors: incorporating $q > 0$ lowers a call value; it improves fit when the zero-dividend benchmark is too high, but can worsen fit when the independently calibrated benchmark already underprices the market.

Table 2: Pricing accuracy using independently estimated pre-sample volatility

Ticker	N	MAE $q = 0$	MAE $q > 0$	NMAE $q = 0$	NMAE $q > 0$	MAE change
AAPL	654	0.2265	0.2269	0.405%	0.406%	-0.15%
BAC	388	0.0756	0.0727	1.230%	1.182%	+3.89%
JPM	465	0.4414	0.5159	1.195%	1.396%	-16.88%
MRK	270	0.1762	0.1987	3.306%	3.727%	-12.74%
MSFT	786	0.4156	0.4473	0.809%	0.870%	-7.62%
NVDA	1336	0.3190	0.3192	0.625%	0.625%	-0.07%
PFE	189	0.0565	0.0588	2.647%	2.755%	-4.11%
UNH	700	1.2305	1.1741	3.171%	3.025%	+4.58%
WFC	327	0.1146	0.1292	1.781%	2.007%	-12.69%

Notes: MAE change is $100(MAE_0 - MAE_q)/MAE_0$. Positive values indicate improved accuracy. NMAE is normalized by the sum of observed market midpoints within ticker.

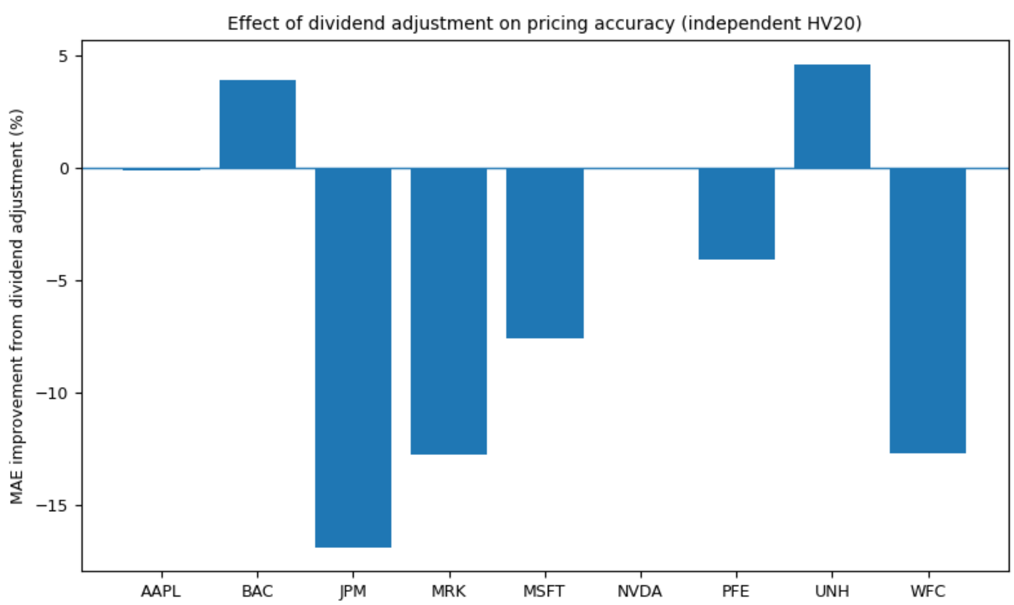

Figure 3: Percentage change in ticker-level MAE after incorporating the dividend yield under the independent-volatility specification. Positive values indicate lower error.

Table 3 reports inference on ticker-day averages. At the individual-ticker level, the UNH improvement is statistically detectable under both tests and occurs on 11 of 13 sample days. JPM and WFC are worse on all 13 days; MRK and MSFT also show statistically significant deterioration. AAPL, BAC, and PFE do not show a statistically detectable daily difference. NVDA's difference is statistically detectable by some tests but economically negligible, with pooled MAE changing by only 0.07%. These tests are descriptive given the 13-day window, and the table reports unadjusted p-values. Under a Holm familywise correction across nine ticker comparisons, the large deteriorations remain significant, whereas the UNH improvement no longer meets a 5% familywise threshold (adjusted $p = 0.063$ for the paired t-test and $p = 0.054$ for Wilcoxon). Statistical and economic significance are therefore reported separately.

Table 3: Paired daily tests of the dividend adjustment

Ticker	Mean daily $MAE_0 - MAE_q$	Paired t p	Wilcoxon p	Days $q > 0$ better
AAPL	-0.0004	0.816	0.635	5/13
BAC	+0.0020	0.705	0.787	4/13
JPM	-0.0777	< 0.001	< 0.001	0/13
MRK	-0.0239	0.003	0.005	1/13
MSFT	-0.0330	< 0.001	< 0.001	1/13
NVDA	-0.0002	0.018	< 0.001	1/13
PFE	-0.0021	0.396	0.414	5/13
UNH	+0.0633	0.013	0.013	11/13
WFC	-0.0144	< 0.001	< 0.001	0/13

Notes: Each ticker contributes 13 paired daily MAEs. A positive mean difference indicates lower error under the dividend-adjusted specification.

4.4 American exercise-style diagnostic

The 100-step CRR diagnostic indicates that the early-exercise component is small on average for these short-maturity calls: the mean American-minus-European tree premium is below one cent for every ticker. The broad cross-sectional conclusion is therefore not driven by exercise style, although small borderline cases can change. Most notably, PFE's European dividend-adjusted MAE is slightly worse than the $q = 0$ benchmark, whereas the American dividend-paying tree produces a small improvement. This confirms that residual market–

model differences should not be attributed exclusively to dividends.

Table 4: American-style CRR diagnostic

Ticker	Mean ex. premium	P95	Maximum	American MAE	Change vs $q = 0$
AAPL	0.0006	0.0000	0.0474	0.2274	-0.40%
BAC	0.0001	0.0000	0.0097	0.0727	+3.89%
JPM	0.0000	0.0000	0.0000	0.5156	-16.82%
MRK	0.0013	0.0098	0.0332	0.1986	-12.72%
MSFT	0.0000	0.0000	0.0000	0.4473	-7.63%
NVDA	0.0000	0.0000	0.0000	0.3192	-0.08%
PFE	0.0066	0.0241	0.0684	0.0549	+2.84%
UNH	0.0034	0.0217	0.1429	1.1735	+4.63%
WFC	0.0000	0.0000	0.0033	0.1291	-12.62%

Notes: Exercise premium is the 100-step American CRR value minus the corresponding European CRR value using the same S, K, T, r, q, and historical volatility. Positive "change vs $q = 0$" means the American dividend-paying value has lower MAE than the zero-dividend BSM benchmark.

4.5 Robustness and implied-volatility diagnostic

The main patterns are stable when pre-sample volatility is estimated from 10 or 15 returns, when the activity threshold is varied, and when observations with relative spreads above 25%, 50%, or 100% are excluded. Strong effects for JPM, MRK, MSFT, UNH, and WFC retain their direction. The smaller BAC and PFE effects are more sensitive to activity and market-price choices, reinforcing the distinction between economically robust and borderline findings. Using the recent transaction price instead of the midpoint also changes some small effects; this is why the midpoint is the primary market-price proxy and why exact transaction/quote synchronization remains a limitation.

The Yahoo-implied-volatility diagnostic produces a markedly different picture. On the same cleaned sample, the dividend-adjusted model appears to reduce MAE for every ticker, while the correlation between market midpoint and C_q ranges from 0.999476 to 0.999988. Because the volatility input itself contains information from option prices, these near-perfect correlations are not interpreted as independent evidence of pricing accuracy. Their contrast with the independent-volatility results illustrates the circularity concern directly.

5 Discussion and Conclusion

This study distinguishes two questions that are often conflated. First, what is the mechanical BSM price effect of omitting a positive dividend yield? Second, does correcting that omission move an otherwise simplified model closer to observed option prices? The first question has a clear theoretical answer: setting q to zero raises a European call value, and for the short maturities in this sample the first-order dividend sensitivity reproduces the exact price difference almost perfectly. The second question is empirical and does not have a uniform answer.

Once volatility is estimated independently from option premiums and stale or invalid records are screened uniformly, the dividend adjustment improves market-price accuracy for some underlyings but worsens it for others. UNH provides the clearest persistent improvement; BAC shows a positive pooled effect that is not statistically significant at the daily level; AAPL and NVDA are economically close to unchanged; and several underlyings show larger errors after the dividend correction. This does not contradict the BSM dividend formula. It means that the sign of the model correction and the sign of the market-pricing improvement are different objects: lowering a model price helps only when the unadjusted benchmark lies above the relevant market price.

The analysis also clarifies several limitations. The primary volatility measure is deliberately simple and fixed over a short sample, so it should be interpreted as an independent benchmark rather than an optimal volatility forecast. The continuous dividend yield approximates discrete corporate dividends. The observed contracts are American-style, although the CRR diagnostic suggests that average early-exercise premiums are small in this short-maturity sample and highlights PFE as a case where the exercise feature can affect a borderline result. Finally, the downloaded files contain a last-transaction timestamp but not a separate timestamp for the displayed bid and ask, so exact intraday synchronization cannot be established retrospectively.

The practical implication is therefore narrower than a claim that dividend adjustment universally improves BSM accuracy. Dividend omission creates a predictable and readily approximated valuation distortion whose magnitude depends on q, T, S, and moneyness. Whether correcting that distortion improves empirical fit depends on the direction and size of the other model and data discrepancies. This distinction makes the BSM framework useful as a transparent benchmark while avoiding circular validation through option-implied volatility.

Declarations

Ethical approval

Not applicable. This study did not involve human participants, animals, or clinical data requiring ethical review.

Consent to participate

Not applicable.

Consent to publish

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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Author contributions

A.M. developed the core mathematical framework and conducted the empirical data analysis. A.M. and O.I. implemented the computational models and prepared the figures. A.M. and O.I. wrote the manuscript. All authors reviewed and approved the final manuscript.

Data availability

The option-chain inputs used in this study were collected from Yahoo Finance. The original data and computational materials are available at `https://github.com/abdallahmaiga10/BSM-option-pricing`. The revision replication package also contains the exact 13 option snapshots and the pre-sample closing-price series used in the reported analysis.

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